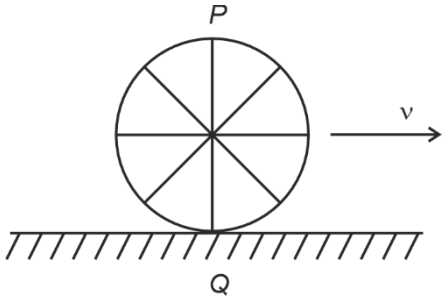
PHYSICS

SECTION-A

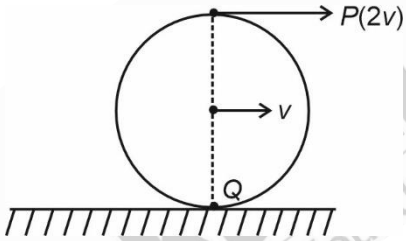
1. A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (*P* and *Q* are any highest and lowest points on the wheel, respectively)?



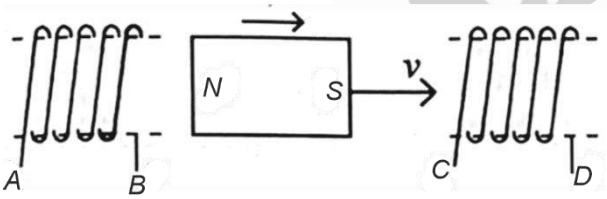
(1) Both the points *P* and *Q* move with equal speed
(2) Point *P* has zero speed
(3) Point *P* moves slower than point *Q*
(4) Point *P* moves faster than point *Q*

Answer (4)

Sol. In the case of pure rolling,



The topmost point will have velocity $2v$ while point *Q* i.e. lowest point will have zero velocity.
Hence point *P* moves faster than point *Q*.

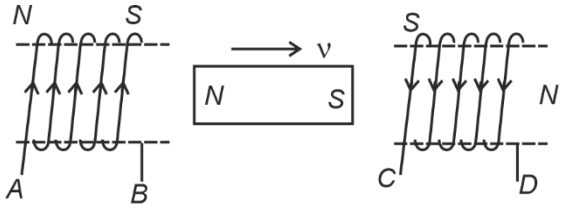
2. 

Solenoid-1 Solenoid-2

In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

(1) *AB* and *CD*
(2) *BA* and *DC*
(3) *AB* and *DC*
(4) *BA* and *CD*

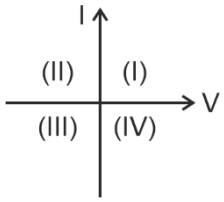
Answer (3)

Sol. 

Solenoid-1 Solenoid-2

North of magnet is moving away from solenoid 1 so end *B* of solenoid 1 is South and as south of magnet is approaching solenoid 2 so end *C* of solenoid 2 is South.

3. Consider the following statements A and B and identify the correct answer:



A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.

B. In a reverse biased *pn* junction diode, the current measured in (μA), is due to majority charge carriers.

(1) Both A and B are correct

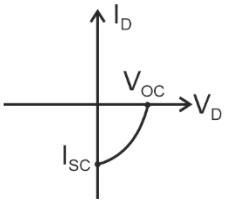
(2) Both A and B are incorrect

(3) A is correct but B is incorrect

(4) A is incorrect but B is correct

Answer (3)

Sol. A: Solar cell characteristics



B: In reverse biased *pn* junction diode, the current measured in (μA), is due to minority charge carrier.

4. A wire of length '*l*' and resistance 100 Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

(1) 55 Ω

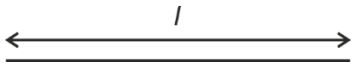
(2) 60 Ω

(3) 26 Ω

(4) 52 Ω

Answer (4)

Sol.



Divided into 10 parts

$$R = \frac{\rho l}{A}$$

$$R' = \frac{\rho l}{10A} = \frac{R}{10}$$

$$R_S = 5 \times \frac{R}{10} \quad \text{[series]}$$

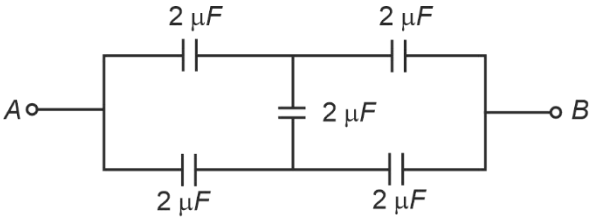
$$R_S = 50$$

$$R_P = \frac{R}{50} \quad \text{[parallel]}$$

$$R_{eq} = R_S + R_P$$

$$= 52\ \Omega$$

5. In the following circuit, the equivalent capacitance between terminal *A* and terminal *B* is :

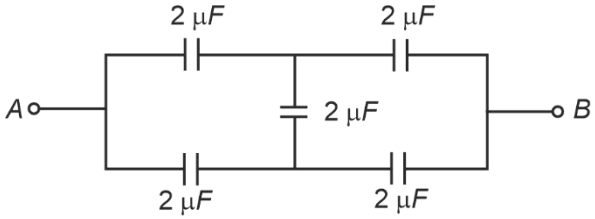


(1) 0.5 μF (2) 4 μF

(3) 2 μF (4) 1 μF

Answer (3)

Sol.



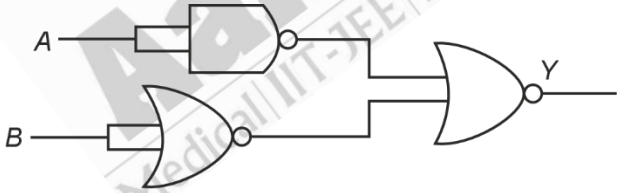
Given circuit is balanced Wheatstone bridge



$C_{AB} = 1 + 1$

$= 2\ \mu F$

6. The output (*Y*) of the given logic gate is similar to the output of an/a

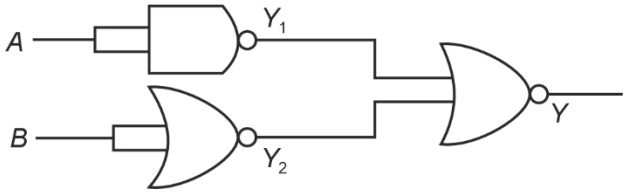


(1) OR gate (2) AND gate

(3) NAND gate (4) NOR gate

Answer (2)

Sol.



$Y_1 = \overline{A \cdot A}$

$= \bar{A}$

$Y_2 = \overline{B + B}$

$= \bar{B}$

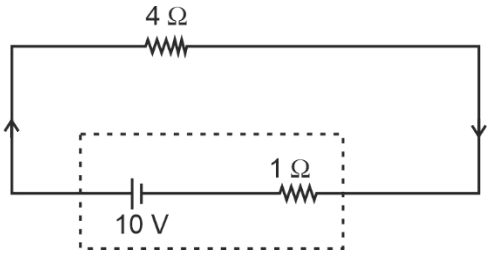
$Y = \overline{Y_1 + Y_2}$

$= \overline{\bar{A} + \bar{B}}$

$= \bar{\bar{A}} \cdot \bar{\bar{B}}$

$= A.B$ is similar to output of AND Gate

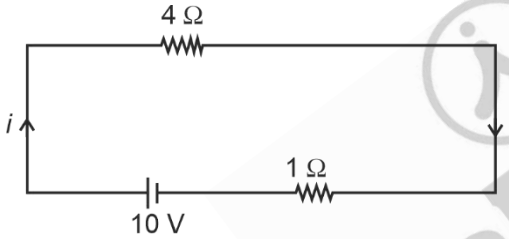
7. The terminal voltage of the battery, whose emf is 10 V and internal resistance 1 Ω, when connected through an external resistance of 4 Ω as shown in the figure is:



(1) 8 V
(2) 10 V
(3) 4 V
(4) 6 V

Answer (1)

Sol.



Current in circuit $i = \frac{10}{4+1} = 2\,A$

Terminal voltage $= E - iR$

$= 10 - 2 \times 1 = 8$ V

8. The mass of a planet is $\frac{1}{10}^{th}$ that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is:

(1) 4.9 m s^{-2}
(2) 3.92 m s^{-2}
(3) 19.6 m s^{-2}
(4) 9.8 m s^{-2}

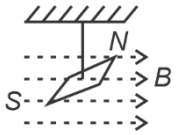
Answer (2)

Sol. $g' = \frac{GM'}{R'^2} = \frac{GM}{10\left(\frac{R}{2}\right)^2}$

$= \frac{4}{10}\frac{GM}{R^2} = 0.4 \times 9.8$

$= 3.92$ m s^{-2}

9. In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is 9.8 x 10^{-6} kg m^2. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}$ Am^2, then the value of 'x' is :



(1) $50\pi^2$ (2) $1280\pi^2$

(3) $5\pi^2$ (4) $128\pi^2$

Answer (2)

Sol. Time period of Oscillation, $T = 2\pi\sqrt{\frac{I}{MB}}$

$$\Rightarrow \frac{1}{4} = 2\pi\sqrt{\frac{9.8\times10^{-6}}{M\times0.049}}$$

$$\Rightarrow \frac{1}{16} = 4\pi^2 \times \frac{9.8\times10^{-6}}{M\times49\times10^{-3}}$$

$$\Rightarrow M = \frac{4\pi^2\times9.8\times10^{-6}}{49\times10^{-3}}\times16$$

$$= \frac{4\pi^2\times9.8\times16\times10^{-3}}{49}$$

$$= 12.8\pi^2 \times 10^{-3} \times 10^{-2} \times 10^2$$

$$= 1280\pi^2 \times 10^{-5}\ Am^2$$

10. If c is the velocity of light in free space, the correct statements about photon among the following are:

A. The energy of a photon is $E = h\nu$.

B. The velocity of a photon is c.

C. The momentum of a photon, $p = \frac{h\nu}{c}$.

D. In a photon-electron collision, both total energy and total momentum are conserved.

E. Photon possesses positive charge.

Choose the correct answer from the options given below:

(1) A, C and D only (2) A, B, D and E only

(3) A and B only (4) A, B, C and D only

Answer (4)

Sol. (A) If c is the velocity of light

so, $E = h\nu$ (Energy of photon)

(B) Velocity of photon is equal to velocity of light i.e. c.

(C) $\lambda = \frac{h}{p}$

$$p = \frac{h}{\lambda}$$

$$p = \frac{h\nu}{c}$$

(D) In photon-electron collision both total energy and total momentum are conserved.

11. The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are 8×10^8 N m^{-2} and 2×10^{11} N m^{-2}, is:

(1) 40 mm (2) 8 mm

(3) 4 mm (4) 0.4 mm

Answer (3)

Sol. In the case for maximum elongation,

Stress = Elastic limit

$$\delta_{max} = \frac{\sigma_{elastic} \times L}{\text{Young's modulus}} = \frac{8\times10^8 \times 1}{2\times10^{11}} = 4\times10^{-3}$$

$$= 4 \text{ mm}$$

i.e. maximum elongation is 4 mm

12. A logic circuit provides the output Y as per the following truth table :

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is :

(1) $\bar{B}$ (2) B

(3) $A.B + \bar{A}$ (4) $A.\bar{B} + \bar{A}$

Answer (1)

Sol.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

According to given truth table, output is independent on value of A

$\therefore$ Output $Y = \bar{B}$

13. The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is 2400 g cm^2. The length of the 400 g rod is nearly:

(1) 20.7 cm (2) 72.0 cm

(3) 8.5 cm (4) 17.5 cm

Answer (3)

Sol. Moment of inertia of rod = $I = \frac{m\ell^2}{12}$

$$\Rightarrow 2400 = 400\frac{\ell^2}{12}$$

$$\Rightarrow 72 = \ell^2$$

$$\Rightarrow \ell = \sqrt{72} = 8.48 \text{ cm} \simeq 8.5 \text{ cm}$$

14. In a vernier callipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:

(1) $100N$ (2) $10(N+1)$

(3) $\frac{1}{10N}$ (4) $\frac{1}{100(N+1)}$

Answer (4)

Sol. $V.C = \text{MSD} - \text{VSD}$ ...(1)

given : $(N+1)$ VSD = N MSD

$$\text{VSD} = \left(\frac{N}{N+1}\right) \text{MSD} \quad ...(2)$$

From (1) and (2)

$$V.C = (\text{MSD}) - \frac{N}{N+1}(\text{MSD})$$

$$= \text{MSD}\left(1 - \frac{N}{N+1}\right) = \frac{\text{MSD}}{N+1}$$

$$= \frac{0.01}{N+1} = \frac{1}{100(N+1)}$$

15. Match List I with List II.

	List I (Spectral Lines of Hydrogen for transitions from)		**List II (Wavelengths (nm))**
A.	$n_2 = 3$ to $n_1 = 2$	I.	410.2
B.	$n_2 = 4$ to $n_1 = 2$	II.	434.1
C.	$n_2 = 5$ to $n_1 = 2$	III.	656.3
D.	$n_2 = 6$ to $n_1 = 2$	IV.	486.1

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-I, D-II (2) A-I, B-II, C-III, D-IV

(3) A-II, B-I, C-IV, D-III (4) A-III, B-IV, C-II, D-I

Answer (4)

Sol. Energy difference $\Delta E = \frac{hc}{\lambda}$

$$\therefore \lambda \propto \frac{1}{\Delta E}$$

$$(\Delta E)_{6-2} > (\Delta E)_{5-2} > (\Delta E)_{4-2} > (\Delta E)_{3-2}$$

$$\lambda_{6-2} < \lambda_{5-2} < \lambda_{4-2} < \lambda_{3-2}$$

A-III, B-IV, C-II, D-I

16. If the monochromatic source in Young's double slit experiment is replaced by white light, then

(1) There will be a central bright white fringe surrounded by a few coloured fringes

(2) All bright fringes will be of equal width

(3) Interference pattern will disappear

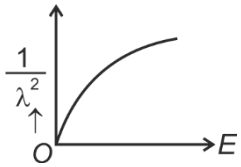
(4) There will be a central dark fringe surrounded by a few coloured fringes

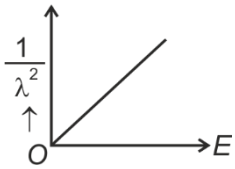
Answer (1)

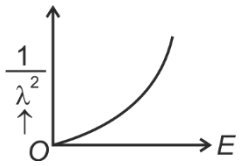
Sol. At central point on screen, path difference is zero for all wavelength. So, central bright fringe is white and other fringes depend on wavelength as $\beta = \frac{\lambda D}{d}$.

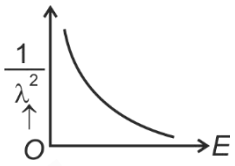
Therefore, other fringes will be coloured.

17. The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle):

(1) 

(2) 

(3) 

(4) 

Answer (2)

Sol. de-Broglie wavelength $\lambda = \frac{h}{P} = \frac{h}{mv} = \frac{h}{\sqrt{2\,mE}}$ where $E = \frac{1}{2}mv^2$

Squaring both sides,

$\lambda^2 = \frac{h^2}{4m^2E}$

$\Rightarrow \frac{1}{\lambda^2} = (\text{constant})\,E$

Graph passes through origin with constant slope.

18. Match **List-I** with **List-II**.

	List-I (Material)		**List-II (Susceptibility (χ))**
A.	Diamagnetic	I.	$\chi = 0$
B.	Ferromagnetic	II.	$0 > \chi \geq -1$
C.	Paramagnetic	III.	$\chi >> 1$
D.	Non-magnetic	IV.	$0 < \chi < \varepsilon$ (a small positive number)

Choose the correct answer from the options given below

(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-III, D-IV

Answer (3)

Sol.

(Material)		**(Susceptibility (χ))**
Diamagnetic	(II)	$0 > \chi \geq -1$
Ferromagnetic	(III)	$\chi >> 1$
Paramagnetic	(IV)	$0 < \chi < \varepsilon$
Non-magnetic	(I)	$\chi = 0$

19. An unpolarised light beam strikes a glass surface at Brewster's angle. Then

(1) Both the reflected and refracted light will be completely polarised.

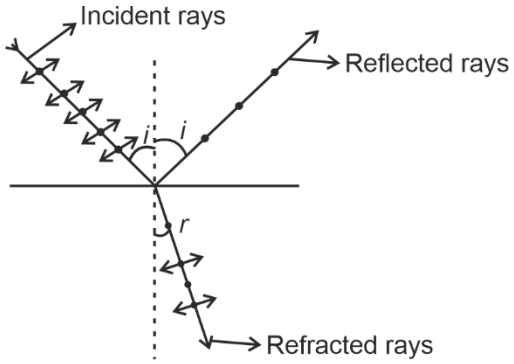
(2) The reflected light will be completely polarised but the refracted light will be partially polarised.

(3) The reflected light will be partially polarised.

(4) The refracted light will be completely polarised.

Answer (2)

Sol.



According to Brewster's law, reflected rays are completely polarized and refracted rays are partially polarized.

20. A particle moving with uniform speed in a circular path maintains:

(1) Constant velocity but varying acceleration

(2) Varying velocity and varying acceleration

(3) Constant velocity

(4) Constant acceleration

Answer (2)

Sol. A particle moving with uniform speed in a circular path maintains varying velocity and varying acceleration. It is because direction of both velocity as well as acceleration will change continuously.

21. The quantities which have the same dimensions as those of solid angle are:

(1) strain and arc

(2) angular speed and stress

(3) strain and angle

(4) stress and angle

Answer (3)

Sol. Solid angle $d\Omega = \frac{dA}{r^2}$ has dimensions $[M^0L^0T^0]$

Strain $= \frac{\Delta l}{l}$ has dimensions $[M^0L^0T^0]$

Angle measured in radians is also dimensionless $[M^0L^0T^0]$

$\theta = \frac{l}{r}$

22. A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2ω while keeping the same radius, the tension in the string becomes:

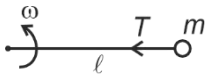
(1) $\frac{T}{4}$

(2) $\sqrt{2}T$

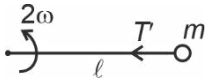
(3) T

(4) $4T$

Answer (4)

Sol. 

$T = m\ell\omega^2$



$T' = m\ell(2\omega)^2$

$T' = 4T$

23. If $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ m represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are

(1) 5 cm, 1 s
(2) 5 m, 1 s
(3) 5 cm, 2 s
(4) 5 m, 2 s

Answer (4)

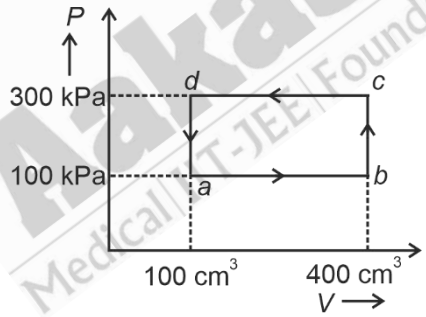
Sol. $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ m

Amplitude = 5 m

$\omega = \pi = \frac{2\pi}{T}$

$T = \frac{2\pi}{\pi} = 2$ s

24. A thermodynamic system is taken through the cycle *abcda*. The work done by the gas along the path *bc* is:



(1) –90 J
(2) –60 J
(3) Zero
(4) 30 J

Answer (3)

Sol. Path *bc* is an isochoric process.

∴ Work done by gas along path *bc* is zero.

25. Given below are two statements:

Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II: Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Statement I is correct but Statement II is incorrect
(2) Statement I is incorrect but Statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

Answer (1)

Sol. Statement I is true as atoms are electrically neutral because they contain equal number of positive and negative charges.

Statement II is wrong as atom of most of the elements are stable and emit characteristic spectrum. But this statement is not true for every atom.

26. In an ideal transformer, the turns ratio is $\frac{N_P}{N_S} = \frac{1}{2}$. The ratio $V_S : V_P$ is equal to (the symbols carry their usual meaning) :

(1) 1 : 1 (2) 1 : 4

(3) 1 : 2 (4) 2 : 1

Answer (4)

Sol. According to transformer ratio,

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = 2:1$$

27. Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is

(1) 4 : 1 (2) 1 : 4

(3) 1 : 2 (4) 2 : 1

Answer (4)

Sol. Before collision $\Rightarrow$ $\overset{A}{\bigcirc} \rightarrow v_1$ $\overset{B}{\bigcirc}$ rest

It undergoes completely inelastic collision

Using conservation of linear momentum

Initial momentum = Final momentum

$\Rightarrow mv_1 = mv_2 + mv_2$

$\Rightarrow mv_1 = 2mv_2$

$\Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$

28. $${}^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$$

In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are

(1) 288, 82

(2) 286, 81

(3) 280, 81

(4) 286, 80

Answer (2)

Sol. $${}^{290}_{82}X \xrightarrow{\alpha} {}^{286}_{80}Y \xrightarrow{e^+} {}^{286}_{79}Z \xrightarrow{\beta^-} {}^{286}_{80}P \xrightarrow{e^-} {}^{286}_{81}Q$$

$A \rightarrow 286$

$Z = 81$

29. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude, 4×10^{-6} C m, is $\pm 9 \times 10^3$ V.

(Take $\frac{1}{4\pi \in_0} = 9 \times 10^9$ SI units)

Reason R: $V = \pm \frac{2P}{4\pi \in_0 r^2}$, where r is the distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

Answer (1)

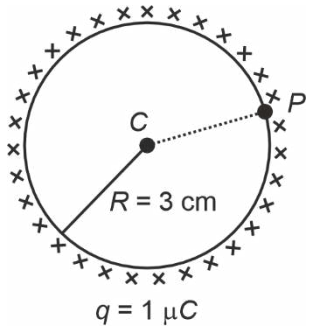
Sol. The potential V at any point, at distance r from centre of dipole $= \frac{KP\cos\theta}{r^2}$

At axial point where $\theta = 0°$, $V = \frac{KP}{r^2} = \frac{9 \times 10^9 \times 4 \times 10^{-6}}{2^2} = 9 \times 10^3$ V

At axial point where $\theta = 180°$, $V = \frac{-KP}{r^2} = -9 \times 10^3$ V

30. A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is:

(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ SI units)



(1) 0.5×10^5

(2) Zero

(3) 3×10^5

(4) 1×10^5

Answer (2)

Sol. For uniformly charged spherical shell,

$V = \frac{kq}{R}$ (For $r \le R$)

$\therefore$ $V_C = V_P$

$V_C - V_P =$ Zero

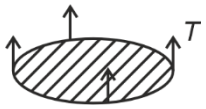
31. A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 N m^{-1}, then the excess force required to take it away from the surface is

(1) 1.98 mN (2) 99 N

(3) 19.8 mN (4) 198 N

Answer (3)

Sol.



Excess force = $T \times 2\pi R$

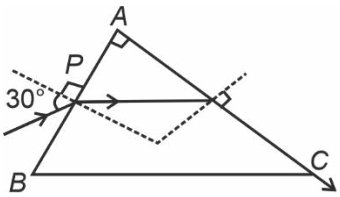
$= \frac{7}{100} \times 2 \times 3.14 \times \frac{4.5}{100}$

$= 197.82 \times 10^{-4}$

$= 19.8 \times 10^{-3}$ N

= 19.8 mN

32. A light ray enters through a right angled prism at point *P* with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base *BC* and emerges along the face *AC*. The refractive index of the prism is:

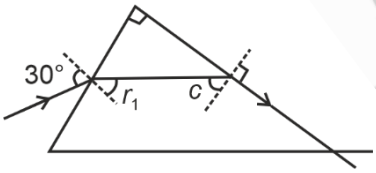


(1) $\frac{\sqrt{3}}{4}$ (2) $\frac{\sqrt{3}}{2}$

(3) $\frac{\sqrt{5}}{4}$ (4) $\frac{\sqrt{5}}{2}$

Answer (4)

Sol.



$A = 90°$

In prism, $r_1 + c = A$

$r_1 = 90° - c$ ...(1)

$\sin c = \frac{1}{\mu} \Rightarrow \cos c = \frac{\sqrt{\mu^2 - 1}}{\mu}$

$\Rightarrow$ Apply Snell's law, on incidence surface

$1 \cdot \sin 30° = \mu \sin(r_1) \Rightarrow 1 \times \frac{1}{2} = \mu \times \sin(90° - c)$

$\frac{1}{2} = \mu \times \frac{\sqrt{\mu^2 - 1}}{\mu}$

On squaring $\frac{1}{4} = \mu^2 - 1$

$\Rightarrow \mu^2 = \frac{5}{4} \Rightarrow \mu = \frac{\sqrt{5}}{2}$

33. At any instant of time t, the displacement of any particle is given by $2t - 1$ (SI unit) under the influence of force of 5 N. The value of instantaneous power is (in SI unit):

(1) 7 (2) 6

(3) 10 (4) 5

Answer (3)

Sol. $x = 2t - 1$

$$v = \frac{dx}{dt} = 2 \text{ m s}^{-1}$$

$P = F.\ v$

$= 2 \times 5 = 10$ W

34. A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

(1) 4.4 mT (2) 44 T

(3) 44 mT (4) 4.4 T

Answer (1)

Sol. The magnitude of magnetic field due to circular coil of N turns is given by

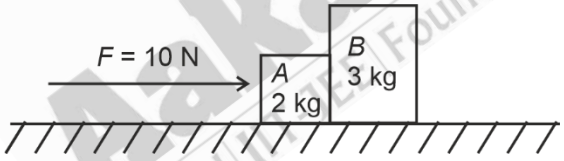
$$B_C = \frac{\mu_0 iN}{2R}$$

$$= \frac{4\pi \times 10^{-7} \times 7 \times 100}{2 \times 0.1}$$

$= 4.4 \times 10^{-3}$ T

$= 4.4$ mT

35. A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is :



(1) 6 N (2) 10 N

(3) Zero (4) 4 N

Answer (1)

Sol. $F = (M_1 + M_2)a$

$$a = \frac{10}{2+3} = 2 \text{ ms}^{-2}$$

$F' = M_2(2) = 3 \times 2$ N $= 6$ N

SECTION-B

36. A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is:

(1) 17 (2) 32

(3) 34 (4) 28

Answer (4)

Sol. $f_0 = 140$ cm and $f_e = 5$ cm

For distant object,

$$m = \frac{f_0}{f_e} = \frac{140}{5} = 28$$

37. Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

(1) 1 : 2 (2) 2 : 3

(3) 1 : 1 (4) 2 : 9

Answer (4)

Sol. Power Consumed = $P = \frac{V^2}{R}$

$\frac{P_A}{P_B} = \frac{R_B}{R_A}$

$R_A = 2R_B$

For Series Combination

$P_S = \frac{V^2}{3R_B}$

For Parallel Combination

$P_P = \frac{3V^2}{2R_B}$

$\frac{P_S}{P_P} = \frac{2}{9}$

38. If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

(1) $2\sqrt{3}$ (2) 4

(3) $\sqrt{3}$ (4) $\sqrt{2}$

Answer (4)

Sol. $T' = 2\pi\sqrt{\frac{\ell'}{g}}$ where $\ell' = \frac{\ell}{2}$

$T = 2\pi\sqrt{\frac{\ell}{g}}$

$T' = \frac{x}{2}T$

$2\pi\sqrt{\frac{\ell}{2g}} = \frac{x}{2}2\pi\sqrt{\frac{\ell}{g}}$

$\frac{1}{\sqrt{2}} = \frac{x}{2} \Rightarrow x = \sqrt{2}$

39. A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates:

(1) Displacement current of magnitude equal to I flows in a direction opposite to that of I

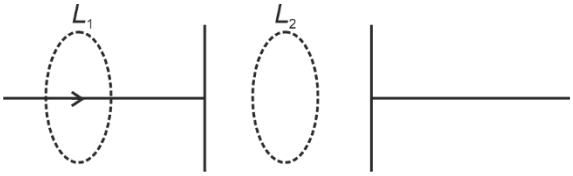
(2) Displacement current of magnitude greater than I flows but can be in any direction

(3) There is no current

(4) Displacement current of magnitude equal to I flows in the same direction as I

Answer (4)

Sol.



According to modified Ampere's law

$\oint B.\,dl = \mu_0(I_C + I_D)$

For Loop L_1 $\quad I_C \neq 0$ and $I_D = 0$

For Loop L_2 $\quad I_C = 0$ and $I_D \neq 0$

Due to KCL $\quad I_C = I_D$

40. A metallic bar of Young's modulus, 0.5×10^{11} N m^{-2} and coefficient of linear thermal expansion 10^{-5} °C^{-1}, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is :

(1) 100×10^3 N (2) 2×10^3 N

(3) 5×10^3 N (4) 50×10^3 N

Answer (4)

Sol. Thermal strain = Longitudinal strain = $\alpha \Delta T$

$\Rightarrow$ Longitudinal strain, $\delta = 10^{-5} \times 10^2 = 10^{-3}$

$\Rightarrow$ Compressive stress = $\delta \times$ Young's Modulus

$= 10^{-3} \times 0.5 \times 10^{11}$

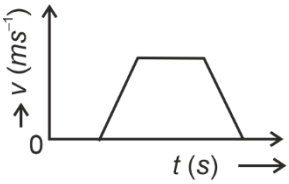
$= 0.5 \times 10^8$

$\Rightarrow$ Compressive force = $0.5 \times 10^8 \times 10^{-3} = 0.5 \times 10^5$

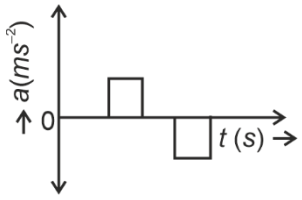
$= 5 \times 10^4 \times \frac{10}{10}$

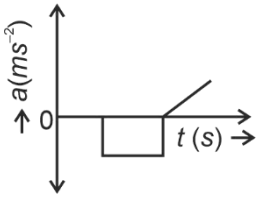
$= 50 \times 10^3$ N

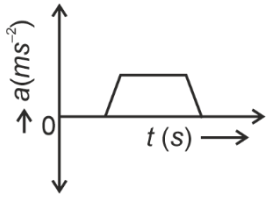
41. The velocity (v) – time (t) plot of the motion of a body is shown below:

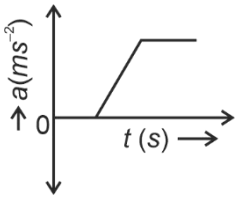


The acceleration (a) – time (t) graph that best suits this motion is :

(1)


(2)


(3)


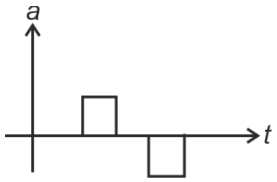
(4)


Answer (1)

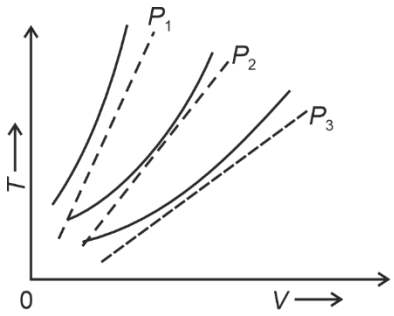
Sol. Initially, the body has zero velocity and zero slope. Hence the acceleration would be zero initially. After that, the slope of v-t curve is constant and positive.

After some time, velocity becomes constant and acceleration is zero.

After that, the slope of v-t curve is constant and negative.



42. The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.



Then the correct relation is:

(1) $P_2 > P_1 > P_3$ (2) $P_1 > P_2 > P_3$

(3) $P_3 > P_2 > P_1$ (4) $P_1 > P_3 > P_2$

Answer (2)

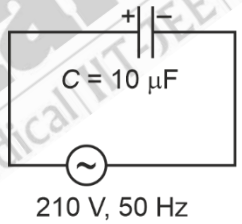
Sol. At same temperature, curve with higher volume corresponds to lower pressure.

$V_3 > V_2 > V_1$

$\Rightarrow P_1 > P_2 > P_3$

(We draw a straight line parallel to volume axis to get this)

43. A 10 μF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$):



(1) 1.20 A (2) 0.35 A

(3) 0.58 A (4) 0.93 A

Answer (4)

Sol. Capacitive Reactance $X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC} = \frac{1}{2\times 3.14\times 50\times 10\times 10^{-6}}$

$= \frac{1000}{3.14}$

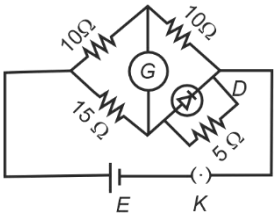
$V_{rms} = 210$ V

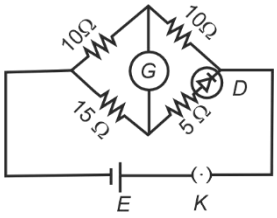
$i_{rms} = \frac{V_{rms}}{X_C} = \frac{210}{X_C}$

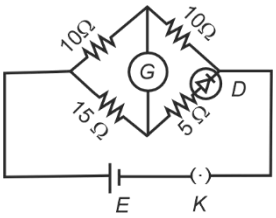
Peak current $= \sqrt{2} i_{rms} = \sqrt{2}\times\frac{210}{1000}\times 3.14 = 0.932$

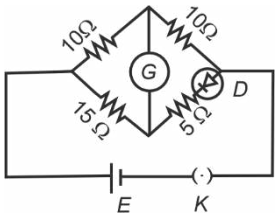
$\simeq 0.93$ A

44. Choose the correct circuit which can achieve the bridge balance.

(1)


(2)


(3)


(4)


Answer (3)

Sol. In option (3),

$$\frac{10}{15}=\frac{10}{5+R_D}$$

The diode can conduct and have resistance $R_D = 10\ \Omega$ because diode have dynamic resistance. In that case bridge will be balanced.

45. The property which is not of an electromagnetic wave travelling in free space is that:

(1) They travel with a speed equal to $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$

(2) They originate from charges moving with uniform speed

(3) They are transverse in nature

(4) The energy density in electric field is equal to energy density in magnetic field

Answer (2)

Sol. The EM waves originate from an accelerating charge. The charge moving with uniform velocity produces steady state magnetic field.

46. The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is:

(1) $\frac{GmM}{2R}$

(2) $\frac{GmM}{3R}$

(3) $\frac{5GmM}{6R}$

(4) $\frac{2GmM}{3R}$

Answer (3)

Sol. Apply energy conservation,

$U_i + K_i = U_f + K_f$

$$\Rightarrow \quad -\frac{GMm}{R}+K_i=-\frac{GMm}{3R}+\frac{1}{2}mv^2$$

$$\Rightarrow \quad -\frac{GMm}{R}+K_i=-\frac{GMm}{3R}+\frac{1}{2}\times m\times\frac{GM}{3R}$$

$$\Rightarrow \quad K_i=-\frac{1}{6}\frac{GMm}{R}+\frac{GMm}{R}$$

$$K_i=\frac{5}{6}\frac{GMm}{R}$$

47. If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then

A. the charge stored in it, increases.
B. the energy stored in it, decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.

Choose the most appropriate answer from the options given below:

(1) B, D and E only
(2) A, B and C only
(3) A, B and E only
(4) A, C and E only

Answer (4)

Sol. Given $V' = V$ = Constant

(i) $C' = \frac{\varepsilon_0 A}{d'}, C = \frac{\varepsilon_0 A}{d}$

$d' < d$

$C' > C$

Hence, final capacitance greater than initial capacitance,

(ii) $U' = \frac{1}{2}C'V^2$

$U = \frac{1}{2}CV^2$

$U' > U$

Hence final energy is greater than initial energy

(iii) $\frac{Q'}{V'} = C'$ and $\frac{Q}{V} = C$

$\frac{Q'}{V'} \neq \frac{Q}{V}$

(iv) Product of charge and voltage

$X' = Q'V = C'V^2$

$X = QV = CV^2$

$X' > X$

48. A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is:

(1) $\alpha\beta t$
(2) $\alpha\beta / t$
(3) $\beta t / \alpha$
(4) $\alpha t / \beta$

Answer (4)

Sol. From principle of homogeneity

$[F] = [\alpha t^2] = [\beta t]$

$[\alpha] = \frac{[F]}{[t^2]}$ and $[\beta] = \frac{[F]}{[t]}$

$\therefore \quad [\alpha]\,[t] = [\beta]$

$\therefore \quad \frac{\alpha t}{\beta} = \text{dimensionless}$

49. A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to:

A. hold the sheet there if it is magnetic.

B. hold the sheet there if it is non-magnetic.

C. move the sheet away from the pole with uniform velocity if it is conducting.

D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

(1) A, C and D only (2) C only

(3) B and D only (4) A and C only

Answer (4)

Sol.

A. A magnetic pole will repel or attract magnetic sheet so force is need.

B. If sheet is non-magnetic, no force needed.

C. If it is conducting, then there will be addy current in sheet, which opposes the motion. So forces is needed move sheet with uniform speed.

D. The non-conducting and non-polar sheet do not interact with magnetic field of magnet.

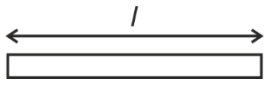
50. An iron bar of length *L* has magnetic moment *M*. It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is :

(1) $2M$ (2) $\frac{M}{\sqrt{3}}$

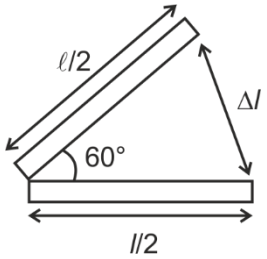
(3) M (4) $\frac{M}{2}$

Answer (4)

Sol.



$M = ml.$



$\Delta l = 2\frac{l}{2}\sin 30°$

$= \frac{l}{2}$

$M' = ml/2$

$= M/2$

CHEMISTRY

SECTION-A

51. Match List I with List II.

List-I (Process)		List-II (Conditions)	
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-III, C-II, D-I

(4) A-IV, B-II, C-III, D-I

Answer (2)

Sol. (A) Isothermal process $\Rightarrow$ Temperature is constant throughout the process

(B) Isochoric process $\Rightarrow$ Volume is constant throughout the process

(C) Isobaric process $\Rightarrow$ Pressure is constant throughout the process

(D) Adiabatic process $\Rightarrow$ No exchange of heat (q) between system and surrounding

52. Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order

n-pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

Answer (3)

Sol. Both statement I and statement II are correct.

Boiling point of n-pentane = 309 K

isopentane = 301 K

neopentane = 282.5

As branching increases molecules attain the shape of a sphere results in smaller area of contact thus weak intermolecular forces between spherical molecules, which are overcome at relatively lower temperature. Leading to decrease in boiling point.